Assignment SQL & OOPS Ticket Booking System

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Task 1: Database Design

1. Create a Database Named 'TicketBookingSystem'

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| hexatraining       |
| information_schema |
| mysql              |
| performance_schema |
| sakila             |
| sys                |
| world              |
+--------------------+
7 rows in set (0.00 sec)

mysql> create database TicketBookingSystem;
Query OK, 1 row affected (0.02 sec)

mysql> use TicketBookingSystem;
Database changed
mysql> |
```

2. Write SQL scripts to create the mentioned tables with appropriate data types, constraints, and relationships.
   a. Venu
   b. Event
   c. Customers
   d. Booking

1. Venue:

```
create table Venue (
    venue_id int(10) AUTO_INCREMENT PRIMARY KEY,
    venue_name varchar(25),
    address varchar(100)
);
```

2. Event:

```
create table Event (
    event_id int AUTO_INCREMENT PRIMARY KEY,
    event_date DATE,
    event_time TIME,
    venue_id int(10),
    total_seats int(100),
    available_seats int(100),
    ticket_price DECIMAL,
    event_type ENUM('Movie', 'Sports', 'Concert') NOT NULL,
    booking_id int(10),
    FOREIGN KEY (venue_id) REFERENCES Venue (venue_id)
);
```

3. Customer:

```
create table customer (
    customer_id int PRIMARY KEY AUTO_INCREMENT,
    email VARCHAR(20) UNIQUE NOT NULL,
    phone_number int(10) UNIQUE NOT NULL
);
```

4. Booking:

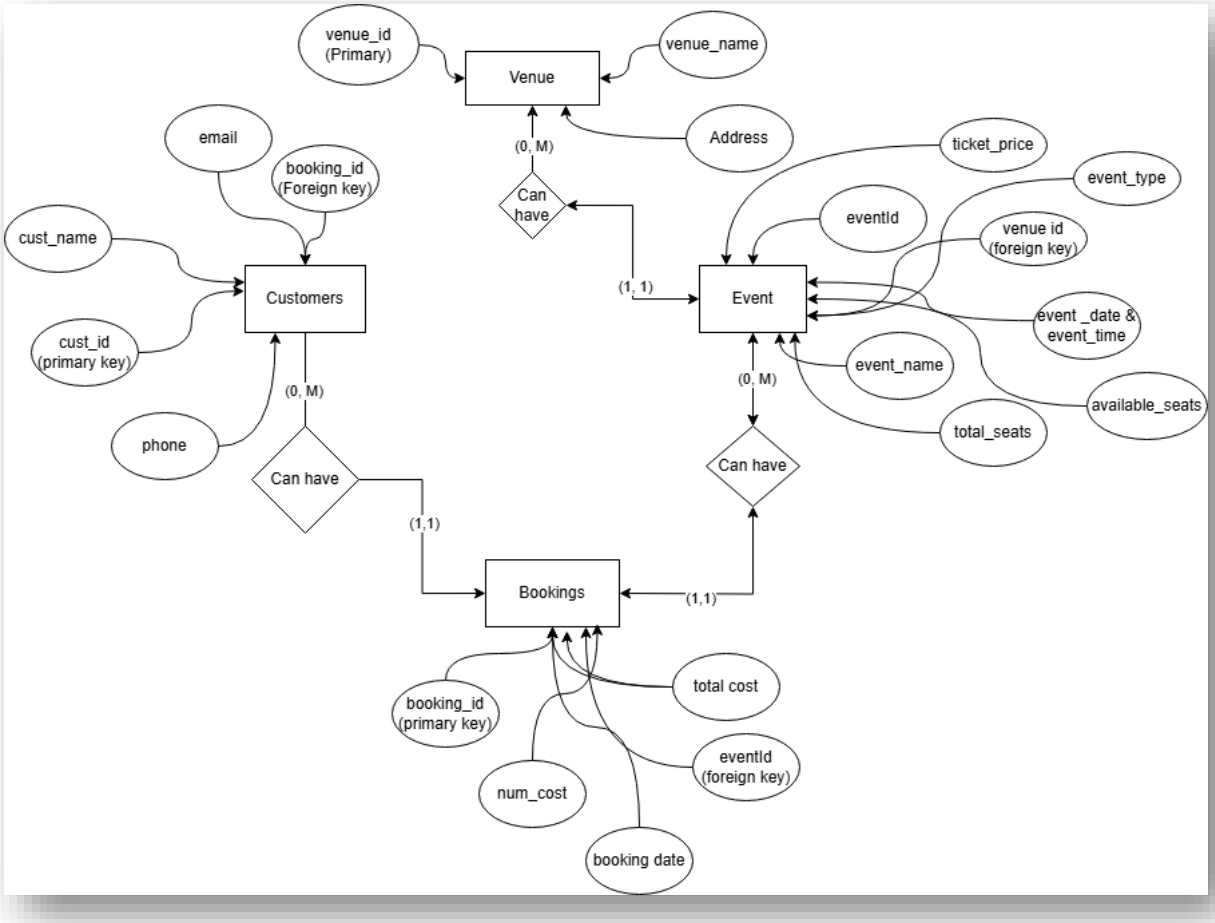
```
create table booking (
    booking_id int PRIMARY KEY AUTO_INCREMENT,
    customer_id int,
    event_id int,
    num_tickets int(50),
    total_cost int(50),
    booking_date DATE,
    FOREIGN KEY (customer_id) REFERENCES customer (customer_id),
    FOREIGN KEY (event_id) REFERENCES Event (event_id)
);
```

3. Creating **ER Diagram** to understand relationships between tables:

   1. **Active elements of the entities of TicketBooking System**
      a. Venue
      b. Event
      c. Customers
      d. Booking
   2. **Relationships:**

a. **Venue ↔ Event:**
    i. (0,M) on Venue means a venue can have zero or many events.
    ii. (1,1) on Event means an event must have exactly one venue.
b. **Event ↔ Booking:**
    i. (0,M) on Event means an event can have zero or many bookings.
    ii. (1,1) on Booking means a booking is for exactly one event.
c. **Customer ↔ Booking:**
    i. (0,M) on Customer means a customer can have zero or many bookings.
    ii. (1,1) on Booking means each booking belongs to exactly one customer.

3. **Finding Primary Key:**
    a. Vid (VenueId)
    b. eventide
    c. customerId
    d. BookingId
4. **Other Attributes**
    a. Venue
        i. Vid (primary key)
        ii. venue_name,
        iii. address
    b. Event
        i. Eventid(primary key)
        ii. event_name,
        iii. event_date DATE,
        iv. event_time TIME,
        v. venue_id (Foreign Key),
        vi. total_seats,
        vii. available_seats,
        viii. ticket_price DECIMAL,
        ix. event_type ('Movie', 'Sports', 'Concert')
        x. booking_id (Foreign Key)
    c. Customer Table
        i. customer_id (Primary key)
        ii. customer_name,
        iii. email,
        iv. phone_number,
        v. booking_id (Foreign Key)
            a.
    d. Booking Table
        i. booking_id (Primary Key),
        ii. customer_id (Foreign Key),
        iii. event_id (Foreign Key),
        iv. num_tickets,
        v. total_cost,
        vi. booking_date,



ER Diagram TicketBookingSystem

4. Create appropriate Primary Key and Foreign Key constraints for referential integrity

```
Alter Table Event add CONSTRAINT FOREIGN KEY(booking_id) REFERENCES
booking(booking_id);
Alter table Customer add column customer_name varchar(255);
Alter table Customer add column booking_id int;
Alter table Customer add CONSTRAINT FOREIGN KEY(booking_id)
REFERENCES booking(booking_id);
```

Using alter command Added NOT NULL Constraint just not to have null values

```
ALTER TABLE Venue
MODIFY venue_name VARCHAR(25) NOT NULL,
MODIFY address VARCHAR(100) NOT NULL;

ALTER TABLE Event
MODIFY event_date DATE NOT NULL,
MODIFY event_time TIME NOT NULL,
MODIFY venue_id INT(10) NOT NULL,
```

```sql
MODIFY total_seats INT(100) NOT NULL,
MODIFY available_seats INT(100) NOT NULL,
MODIFY ticket_price DECIMAL NOT NULL,
MODIFY event_type ENUM('Movie', 'Sports', 'Concert') NOT NULL,
MODIFY booking_id INT(10) NOT NULL;

ALTER TABLE customer
MODIFY email VARCHAR(20) UNIQUE NOT NULL,
MODIFY phone_number INT(10) UNIQUE NOT NULL;

ALTER TABLE booking
MODIFY customer_id INT NOT NULL,
MODIFY event_id INT NOT NULL,
MODIFY num_tickets INT(50) NOT NULL,
MODIFY total_cost INT(50) NOT NULL,
MODIFY booking_date DATE NOT NULL;
```

----------Task 1 Completed----------